UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (November 2026). Answers are unofficial model answers, not official UTM marking schemes.

2. Marking points state the expected answer and the allocation of marks; every question retains its source paper and page reference.

3. Five questions x 20 marks; total = 100 marks.

4. Real-bank questions: 5. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 12 unique ICT450 past papers; every real question remains traceable to its source page.

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This marking scheme consists of 7 printed pages.

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MARKING GUIDE

QUESTION 1 (20 marks)

Source: ICT450-2022-FEB · page [2, 3] · bank id ICT450-2022-FEB-Q1

Reference answer

(a) Four data items in online purchasing. [4 marks]

- Customer identity and contact data.
- Product ID, name, price and quantity.
- Order ID, order date and payment status.
- Delivery address and tracking information.

(b) Relational model features and importance. [4 marks]

- Data is represented as relations (tables) of rows and columns, with keys and integrity constraints.
- End users see a simple tabular view and can query it with SQL without knowing physical storage.
- Designers gain a logical, DBMS-independent structure that supports data independence and controlled redundancy.

(c) Keys and integrity for the academy tables. [12 marks]

- i) Superkeys include {STUDENT_IC}, {STUDENT_ID}, {EMAIL}, {STUDENT_IC, STUDENT_ID} and {STUDENT_ID, EMAIL}.
- Candidate keys: STUDENT_IC, STUDENT_ID and EMAIL, because each is unique and minimal.
- Primary key: STUDENT_ID — it is unique, stable and shorter than the IC.
- ii) STUDENT: entity integrity YES because STUDENT_ID is unique and non-null; referential integrity YES because every COURSE_ID exists in COURSE.
- COURSE: entity integrity YES because COURSE_ID is the unique primary key; referential integrity is not applicable because it has no foreign key.

Marking guide

- a) 1 mark for each of four data items (4).
- b) 2 marks for the features and 2 marks for importance (4).
- c)(i) 2 marks for superkeys, 2 for candidate keys and 2 for the primary key/reason (6).
- c)(ii) 3 marks for each table's integrity explanation (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 1 DATABASE CONCEPT (1).pdf; TOPIC 2 DATA MODEL (1).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- The STUDENT table has unique STUDENT_IC, STUDENT_ID and EMAIL values; the third table is the COURSE table shown.

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QUESTION 2 (20 marks)

Source: ICT450-2023-JUL · page [2] · bank id ICT450-2023-JUL-Q2

Reference answer

(a) Four activities in the database design phase. [8 marks]

- Create the conceptual design — identify entities, attributes, relationships and business rules.
- Select the DBMS software — choose a platform that fits the data model, cost and hardware.
- Create the logical design — map the conceptual model to tables, keys, domains and constraints.
- Create the physical design — choose storage, indexes, file organisation and access paths.

(b) Mobile database suitability and three usage factors. [8 marks]

- Yes, a mobile database is suitable for a company with branches in Malaysia and Indonesia because staff need access away from the office.
- Factor 1: ubiquitous connectivity — field staff can reach the database from any branch or location.
- Factor 2: synchronization — local device data can be reconciled with the central database when connectivity returns.
- Factor 3: portability and location-aware services — mobile devices support on-site inventory and customer service.

(c) Structured versus unstructured data. [4 marks]

- Structured data has a predefined schema and is stored in rows and columns; example: a sales table.
- Unstructured data has no predefined relational schema; example: images, video, audio or free-text documents.

Marking guide

- a) 2 marks for each of four design activities (8).
- b) 2 marks for suitability and 2 marks for each of three factors (2 + 6 = 8).
- c) 2 marks for each data type with example (4).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 2 DATA MODEL (1).pdf; TOPIC 6 DATABASE DESIGN (1).pdf

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QUESTION 3 (20 marks)

Source: ICT450-2023-FEB · page [4] · bank id ICT450-2023-FEB-Q3

Reference answer

The boarding schema contains a repeating group of flight and destination attributes for each customer booking.

(a) Define normalization and its importance. [3 marks]

- Normalization is the process of evaluating and correcting table structures to minimise redundancy and anomalies.
- It is important because it prevents insertion, update and deletion anomalies and keeps each fact in one place.

(b) Three requirements for good normalized tables. [6 marks]

- Each cell must be atomic and there must be no repeating groups; example: one row per boarding pass, not a list of flights in one cell.
- Every non-key attribute must depend on the whole primary key; example: customer name depends on CustIC, not on FlightID.
- There must be no transitive dependency; example: DestinationName must depend on DestinationID, not on FlightID through the destination.

(c) Normalise to 3NF. [11 marks]

1NF: expand the repeating group to one row per customer-flight-date; key (CustIC, FlightID, DateDepart).

2NF: remove customer, flight and destination partial dependencies.

2NF relations.

```
CUSTOMER (CustIC, CustFirstName, CustLastName)
FLIGHT (FlightID, PlaneType, Capacity, DestinationID)
DESTINATION (DestinationID, DestinationName)
BOARDING (CustIC, FlightID, DateDepart, DateArrive, TimeDepart, TimeArrive, Seat)
```

3NF: separate the flight schedule so date/time attributes depend on FlightID and DateDepart, not on the passenger.

3NF relations.

```
CUSTOMER (CustIC, CustFirstName, CustLastName)
DESTINATION (DestinationID, DestinationName)
FLIGHT (FlightID, PlaneType, Capacity, DestinationID)
FLIGHT_SCHEDULE (FlightID, DateDepart, DateArrive, TimeDepart, TimeArrive)
BOARDING (CustIC, FlightID, DateDepart, Seat)
```

Marking guide

- a) 3 marks: definition and importance.
- b) 6 marks: 2 marks for each requirement with example.
- c) 11 marks: 1NF (2), 2NF (4), 3NF (5).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 5 NORMALIZATION (1).pdf

Assumptions / notes

- The curly braces in the printed schema denote a repeating group of flight/destination attributes.

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QUESTION 4 (20 marks)

Source: ICT450-2024-JAN · page [5] · bank id ICT450-2024-JAN-Q4

Reference answer

Movie database schema.

```
MOVIE (MovieID, MovieName, Genre, DateRelease, NetProfit, DirectorID*)
DIRECTOR (DirectorID, DirectorName, Nationality)
REVIEW (ReviewID, Description, MovieID*, ReviewerID*)
REVIEWER (ReviewerID, ReviewerName)
```

(a) Movies released from 2021 until 2023. [4 marks]

```
SELECT *
FROM MOVIE
WHERE DateRelease BETWEEN DATE '2021-01-01' AND DATE '2023-12-31';
```

(b) Total movies directed by Malaysian directors. [4 marks]

```
SELECT COUNT(*) AS total_movies
FROM MOVIE m
JOIN DIRECTOR d ON d.DirectorID = m.DirectorID
WHERE d.Nationality = 'Malaysian';
```

(c) Average net profit for each genre. [4 marks]

```
SELECT Genre, AVG(NetProfit) AS "Average Net Profit"
FROM MOVIE
GROUP BY Genre;
```

(d) Reviewers with more than two reviews. [4 marks]

```
SELECT r.ReviewerName
FROM REVIEWER r
JOIN REVIEW v ON v.ReviewerID = r.ReviewerID
GROUP BY r.ReviewerID, r.ReviewerName
HAVING COUNT(*) > 2;
```

(e) Movie that received the most reviews. [4 marks]

```
SELECT m.MovieName
FROM MOVIE m
JOIN REVIEW v ON v.MovieID = m.MovieID
GROUP BY m.MovieID, m.MovieName
ORDER BY COUNT(*) DESC
FETCH FIRST 1 ROW ONLY;
```

MySQL: replace FETCH FIRST 1 ROW ONLY with LIMIT 1.

Marking guide

- a) 4 marks: date range and SELECT.
- b) 4 marks: movie-director join and Malaysian filter.
- c) 4 marks: AVG, GROUP BY genre and alias.
- d) 4 marks: join, GROUP BY reviewer, HAVING > 2.
- e) 4 marks: join, COUNT, descending order and top-1.

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 7 SQL.pdf

Assumptions / notes

- DateRelease is stored as a date; 'most reviews' is answered with a top-1 ordered query.

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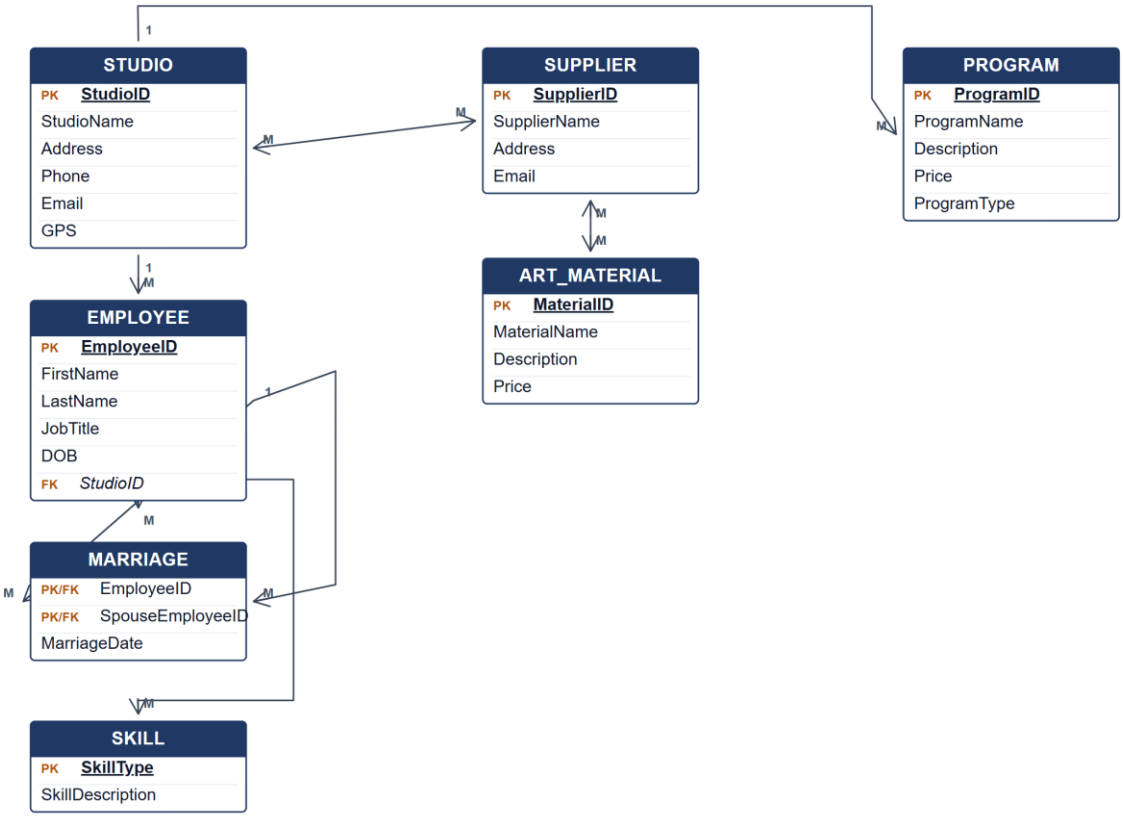
QUESTION 5 (20 marks)

Source: ICT450-2025-FEB · page [5] · bank id ICT450-2025-FEB-Q5

Reference answer

The ERD separates studios, employees, programs, suppliers, materials and skills. The many-to-many relationships are drawn directly between entities. At implementation they become junction tables: STUDIO_SUPPLIER, SUPPLIER_MATERIAL and EMPLOYEE_SKILL. The PROGRAM subtype attributes are listed below the diagram.

(a) Construct the ERD with Crow's Foot notation. [16 marks]



Crow's Foot ERD for the Excellent Art database.

PROGRAM subtype attributes

Program type	Subtype attribute
Visual Art	ClassType
Glass Painting	ProgramLevel (1–5)
Handicraft	CraftData

(b) Two reports from the ERD. [4 marks]

- Studio programme report — list each studio with its programmes, programme type and subtype-specific details such as class type, program level or craft data.
- Supplier and material report — show each studio's suppliers and the art materials each supplier provides, with prices.
- Employee skills report — list employees by studio with their acquired skills.

Marking guide

- a) 16 marks: STUDIO, EMPLOYEE, PROGRAM and SUPPLIER (4); ART_MATERIAL, SKILL and MARRIAGE (3); PROGRAM subtype attributes (2); direct many-to-many relationships and their junction tables (3); keys and Crow's Foot cardinalities (4).
- b) 2 marks for each of two relevant reports explained (2 × 2 = 4).

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Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 4 ERD_stu (2).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- PROGRAM subtypes share the ProgramID key. Their subtype attributes are listed below the diagram rather than drawn as separate boxes, keeping the ERD readable.
- A marriage is represented as an associative entity linking two EMPLOYEE rows, because it has a marriage date and involves two employees.
- STUDIO deals with many SUPPLIERs, and SUPPLIER supplies many ART_MATERIALs; both many-to-many relationships are drawn directly and become junction tables at implementation.

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